STAT 8020 R Session 10: Randomized Complete Block Designs, Factorial Designs, and Split-Plot Designs

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Contents

Randomized Complete Block Design (RCBD)

In a Randomized Complete Block Design (RCBD), each block contains all treatments exactly once. Blocking is used when experimental units can be grouped into relatively homogeneous blocks. The goal is to compare treatment effects while accounting for block-to-block variation.

In this example covered in lecture, we compare three treatments, labeled A, B, and C, across five blocks.

Create the data set

```r
# Response values
# These are the observed outcomes from the experiment
x <- c(52, 47, 44, 51, 42,
       60, 55, 49, 52, 43,
       56, 48, 45, 44, 38)

# Treatment labels
# There are 3 treatments: A, B, and C
# Each treatment appears once in each of the 5 blocks
trt <- rep(c("A", "B", "C"), each = 5)

# Block labels
# There are 5 blocks, repeated for each treatment
blk <- rep(1:5, 3)

# Create a data frame for analysis
# Convert block to a factor because blocks are categorical
dat <- data.frame(x = x,
  trt = as.factor(trt),
  blk = as.factor(blk))

# Display the dataset
dat
```

```
##     x trt blk
## 1  52   A   1
## 2  47   A   2
## 3  44   A   3
## 4  51   A   4
## 5  42   A   5
## 6  60   B   1
## 7  55   B   2
## 8  49   B   3
## 9  52   B   4
## 10 43   B   5
## 11 56   C   1
## 12 48   C   2
## 13 45   C   3
## 14 44   C   4
## 15 38   C   5
```

Fit the Model

For an RCBD, the additive model is

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij},$$

where:

- μ is the baseline mean,

- τ_i is the effect of treatment i,
- β_j is the effect of block j,
- ε_{ij} is the random error term.

This model adjusts for block effects when comparing treatments.

```r
# Fit the RCBD model
# trt represents treatment effects. blk represents block effects
rcbd_model <- lm(x ~ trt + blk, data = dat)

# Display the ANOVA table for the RCBD model
anova(rcbd_model)
```

```
## Analysis of Variance Table
##
## Response: x
##           Df Sum Sq Mean Sq F value    Pr(>F)
## trt        2   89.2   44.60  7.6239 0.0140226 *
## blk        4  363.6   90.90 15.5385 0.0007684 ***
## Residuals  8   46.8    5.85
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Estimated coefficients
coef(rcbd_model)
```

```
## (Intercept)        trtB        trtC        blk2        blk3        blk4
##        54.8         4.6        -1.0        -6.0       -10.0        -7.0
##        blk5
##       -15.0
```

Interpretation

The treatment row tests whether the treatment means differ after accounting for block effects.

The block row tests whether there are systematic differences among blocks.

Including block effects can reduce the residual error if the blocks explain meaningful variation in the response.

In R's default treatment coding, the intercept represents the baseline mean. For an RCBD model, the baseline corresponds to the reference treatment level (usually the first treatment level) and the reference block level (usually the first block level).

Thus, the intercept estimates

$$E(Y \mid \text{Treatment} = \text{reference level}, \text{Block} = \text{reference level}).$$

For example, if Treatment A and Block 1 are used as the reference levels, then the intercept represents the estimated mean response for Treatment A in Block 1.

Compare with a One-Way ANOVA Model

A one-way ANOVA ignores the block structure and only compares treatments.

```r
# Fit a one-way ANOVA model that ignores blocks
oneway_model <- lm(x ~ trt, data = dat)

# Display the ANOVA table
anova(oneway_model)
```

```
## Analysis of Variance Table
##
## Response: x
##           Df Sum Sq Mean Sq F value Pr(>F)
## trt        2   89.2    44.6  1.3041 0.3073
## Residuals 12  410.4    34.2
```

Interpretation

The one-way ANOVA treats all observations as if they came from a completely randomized design.

Comparing this model with the RCBD model helps us see the effect of blocking:

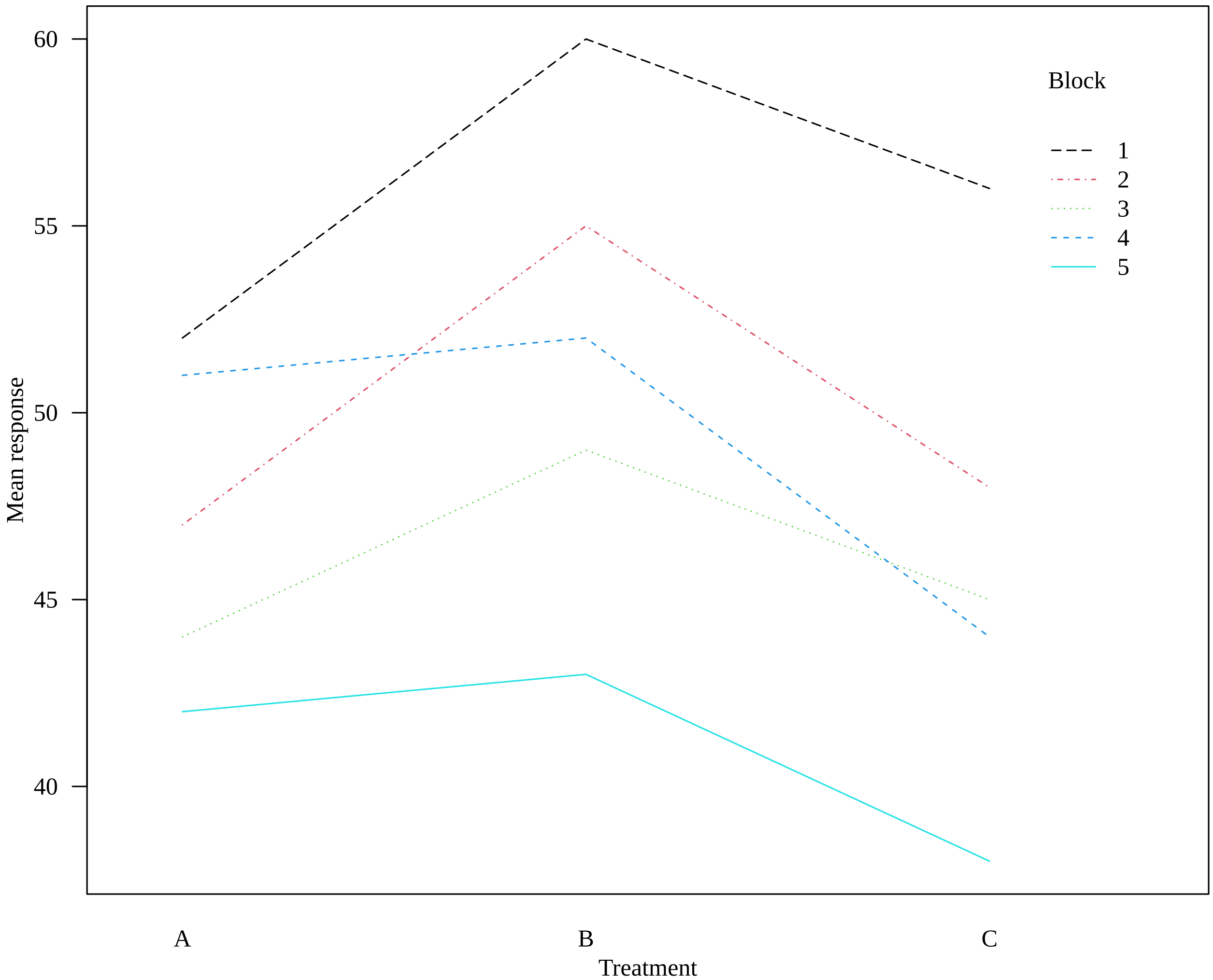
- If blocks explain important variation, the RCBD model will have a smaller residual error.
- A smaller residual error can make the treatment comparison more precise.
- Ignoring blocks may hide treatment effects or lead to less efficient inference.

Interaction Plot: Assessing the Additivity Assumption

In an RCBD, the usual model assumes that treatment effects and block effects are **additive**. This means the treatment differences are assumed to be roughly consistent across blocks.

An interaction plot can help us visually assess whether this assumption is reasonable.

```r
# Create an interaction plot for treatment and block
# The x-axis shows treatment levels
# Separate lines represent different blocks
# Non-parallel lines may suggest possible treatment-by-block interaction
par(mar = c(3.5, 3.5, 1, 0.5), mgp = c(2, 1, 0), las = 1, family = "serif")
interaction.plot(dat$trt, dat$blk, dat$x, col = 1:5,
                 xlab = "Treatment", ylab = "Mean response", trace.label = "Block")
```



Interpretation

If the lines are approximately parallel, the additivity assumption is more reasonable.

If the lines cross frequently or show very different patterns, this may suggest a treatment-by-block interaction. In that case, the simple additive RCBD model may not fully describe the data.

Factorial Design

A factorial design studies two or more factors simultaneously. This allows us to examine:

- the main effect of each factor,
- and the interaction between factors.

In this example, we study how **temperature** and **material type** affect average life.

Create the Dataset

```r
# Response variable: observed lifetime values
y <- c(130, 74, 150, 159, 138, 168, 155, 180, 188, 126, 110, 160,
       34, 80, 136, 106, 174, 150, 40, 75, 122, 115, 120, 139,
       20, 82, 25, 58, 96, 82, 70, 58, 70, 45, 104, 60)

# Factor A: temperature levels
# There are 3 temperature settings: 15, 70, and 125 degrees Fahrenheit
# Each temperature level has 12 observations
temp <- c(rep(15, 12), rep(70, 12), rep(125, 12))

# Factor B: material type
# There are 3 material types
# Each material appears twice within each temperature-material combination
material <- rep(c(1, 1, 2, 2, 3, 3), 6)

# Create the data frame
dat <- data.frame(y = y, temp = as.factor(temp), material = as.factor(material))
```

Compute Cell Means and Marginal Means

We compute the average response by temperature, by material, and by each temperature-material combination.

```r
# Mean response for each temperature level
(meanA <- tapply(dat$y, dat$temp, mean))
```

```
##        15        70       125
## 144.83333 107.58333  64.16667
```

```r
# Mean response for each material type
(meanB <- tapply(dat$y, dat$material, mean))
```

```
##         1         2         3
##  83.16667 108.33333 125.08333
```

```r
# Mean response for each temperature-material combination
(meanAB <- tapply(dat$y,list(dat$temp, dat$material), mean))
```

```
##          1      2      3
## 15  134.75 155.75 144.00
## 70   57.25 119.75 145.75
## 125  57.50  49.50  85.50
```

Interpretation

- `meanA` summarizes the main effect pattern for temperature.
- `meanB` summarizes the main effect pattern for material.
- `meanAB` summarizes the joint effect of temperature and material.

The cell means are especially useful for understanding possible interaction effects.

Two-Way ANOVA with Interaction

For a two-factor factorial design, the full model is

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk},$$

where:

- μ is the overall mean,
- α_i is the effect of temperature level i,
- β_j is the effect of material type j,
- $(\alpha\beta)_{ij}$ is the interaction effect between temperature and material,
- ε_{ijk} is the random error term.

The interaction term asks whether the effect of temperature depends on the material type.

```r
# Fit a two-way ANOVA model with interaction
# temp * material includes:
#   main effect of temp
#   main effect of material
#   interaction between temp and material
factorial_model <- lm(y ~ temp * material, data = dat)

# Display the ANOVA table
anova(factorial_model)
```

```
## Analysis of Variance Table
##
## Response: y
##               Df Sum Sq Mean Sq F value    Pr(>F)    
## temp           2  39119 19559.4 28.9677 1.909e-07 ***
## material       2  10684  5341.9  7.9114  0.001976 ** 
## temp:material  4   9614  2403.4  3.5595  0.018611 *  
## Residuals     27  18231   675.2                      
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

The ANOVA table tests:

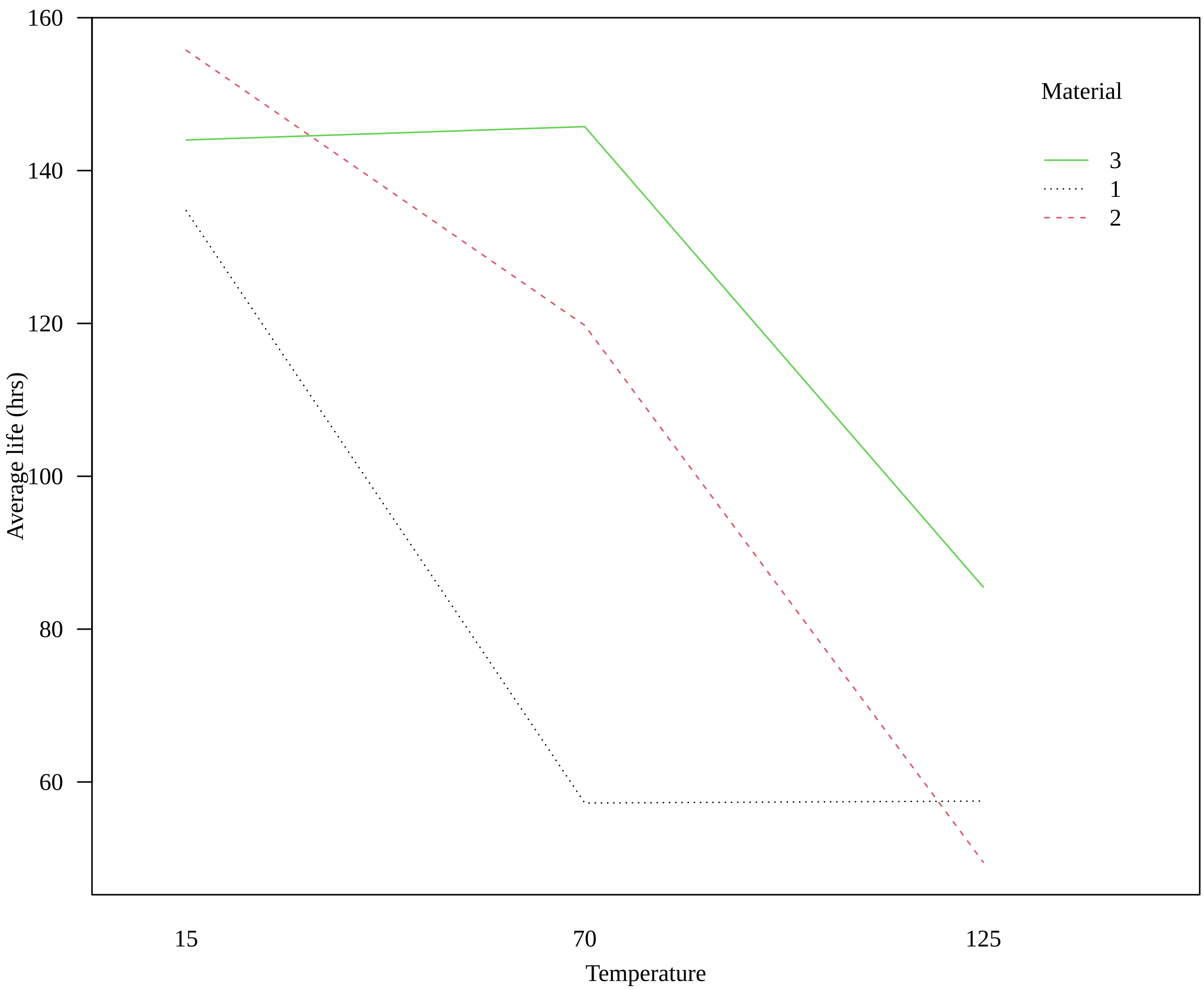
- whether mean lifetime differs by temperature,
- whether mean lifetime differs by material,
- and whether the temperature effect depends on material type.

If the interaction is significant, the main effects should be interpreted carefully because the effect of one factor changes across levels of the other factor.

Interaction Plot

An interaction plot helps visualize whether the temperature effect is similar across materials.

```
# Basic interaction plot
# x-axis: temperature
# separate lines: material types
# response: average lifetime
par(mar = c(3.5, 3.5, 1, 0.5), mgp = c(2.2, 1, 0), las = 1, family = "serif")
interaction.plot(dat$temp, dat$material, dat$y, col = 1:3,
                 xlab = "Temperature", ylab = "Average life (hrs)",
                 trace.label = "Material")
```



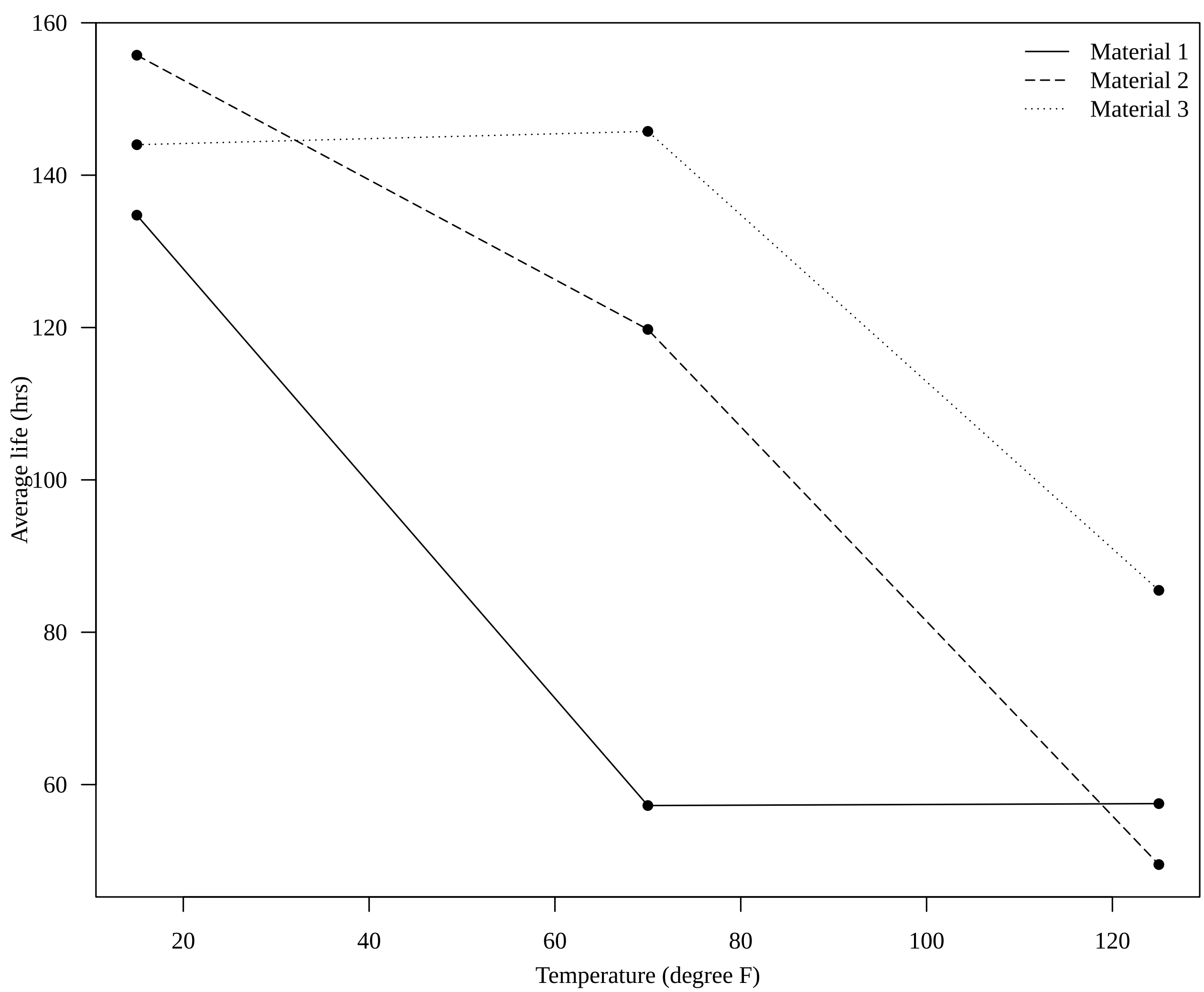
Interpretation

Non-parallel lines suggest that the effect of temperature depends on material type.

Customized Interaction Plot

The customized plot below uses the temperature-material cell means directly.

```r
par(mar = c(3.5, 3.5, 1, 0.5), mgp = c(2.2, 1, 0), las = 1, family = "serif")

# Plot all cell means
plot(rep(c(15, 70, 125), 3), c(meanAB),
     pch = 16, xlab = "Temperature (degree F)", ylab = "Average life (hrs)")

# Add line for Material 1
lines(c(15, 70, 125), meanAB[, 1], lty = 1)

# Add line for Material 2
lines(c(15, 70, 125), meanAB[, 2], lty = 5)

# Add line for Material 3
lines(c(15, 70, 125), meanAB[, 3], lty = 3)

# Add legend
legend("topright", legend = paste("Material", 1:3),
       bty = "n", lty = c(1, 5, 3))
```



Split-Plot Design

This example is adapted from Lukas Meier's ANOVA using `R` notes:

(Link)

A split-plot design is commonly used when some factors are harder or more expensive to randomize than others.

In this example:

- Farmer John has eight large plots of land (the **whole plots**).
- Two fertilization schemes (`control` and `new`) are randomly assigned to the whole plots.
- Each whole plot is then divided into four smaller subplots (the **split plots**).
- Within each whole plot, four strawberry varieties are randomly assigned to the subplots.

The response variable is strawberry fruit mass.

Why a Split-Plot Design?

The key feature is that the two factors are randomized at different experimental-unit levels:

- Fertilizer is randomized at the **whole-plot level**
- Variety is randomized at the **subplot level**

Because of this, observations within the same whole plot are correlated. As a result, the analysis requires a mixed-effects model rather than a standard ANOVA model.

Read the Data

```r
# Read the split-plot dataset
dat <- read.table(
  "http://stat.ethz.ch/~meier/teaching/data/john.dat", header = TRUE)

# Convert plot ID to a factor
# Each plot represents a whole plot
dat[, "plot"] <- factor(dat[, "plot"])

# Display the structure of the dataset
str(dat)
```

```
## 'data.frame':    32 obs. of  4 variables:
##  $ plot      : Factor w/ 8 levels "1","2","3","4",..: 7 7 7 7 5 5 5 5 6 6 ...
##  $ fertilizer: chr  "control" "control" "control" "control" ...
##  $ variety   : chr  "A" "B" "C" "D" ...
##  $ mass      : num  11.6 7.7 12 14 8.9 9.5 11.7 15 10.8 11 ...
```

```r
# View the first few rows
head(dat)
```

```
##   plot fertilizer variety mass
## 1    7    control       A 11.6
## 2    7    control       B  7.7
## 3    7    control       C 12.0
## 4    7    control       D 14.0
## 5    5    control       A  8.9
## 6    5    control       B  9.5
```

Interpretation

The dataset contains:

- `mass`: strawberry fruit mass (response variable)
- `fertilizer`: whole-plot treatment
- `variety`: subplot treatment
- `plot`: whole-plot identifier

The `plot` variable is especially important because observations from the same plot are not independent.

Mixed-Effects Model for Split-Plot Design

A split-plot design typically requires a mixed-effects model because:

- whole plots introduce correlation,
- and different factors are associated with different experimental-unit levels.

The model can be written as

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + u_k + \varepsilon_{ijk},$$

where:

- μ is the overall mean,
- α_i is the fertilizer effect,
- β_j is the variety effect,
- $(\alpha\beta)_{ij}$ is the interaction effect,
- u_k is the random whole-plot effect,
- ε_{ijk} is the subplot error term.

The random plot effect accounts for correlation among observations from the same whole plot.

Fit the Split-Plot Model

```r
# Load the mixed-effects modeling package
library(lmerTest)

# Fit the split-plot mixed-effects model
#
# Fixed effects:
#   fertilizer
#   variety
#   fertilizer-by-variety interaction
#
# Random effect:
#   random intercept for each whole plot
fit <- lmer(mass ~ fertilizer * variety + (1 | plot), data = dat)

# Display the ANOVA table
anova(fit)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                     Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
## fertilizer         137.413 137.413     1     6 68.2395 0.0001702 ***
## variety             96.431  32.144     3    18 15.9627 2.594e-05 ***
## fertilizer:variety   4.172   1.391     3    18  0.6907 0.5695061
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

The ANOVA table tests:

- whether fertilizer affects fruit mass,
- whether strawberry variety affects fruit mass,
- and whether the fertilizer effect depends on variety.

Importantly, the split-plot structure means that:

- fertilizer effects and variety effects use different error structures,
- so they are not tested against the same residual variation.

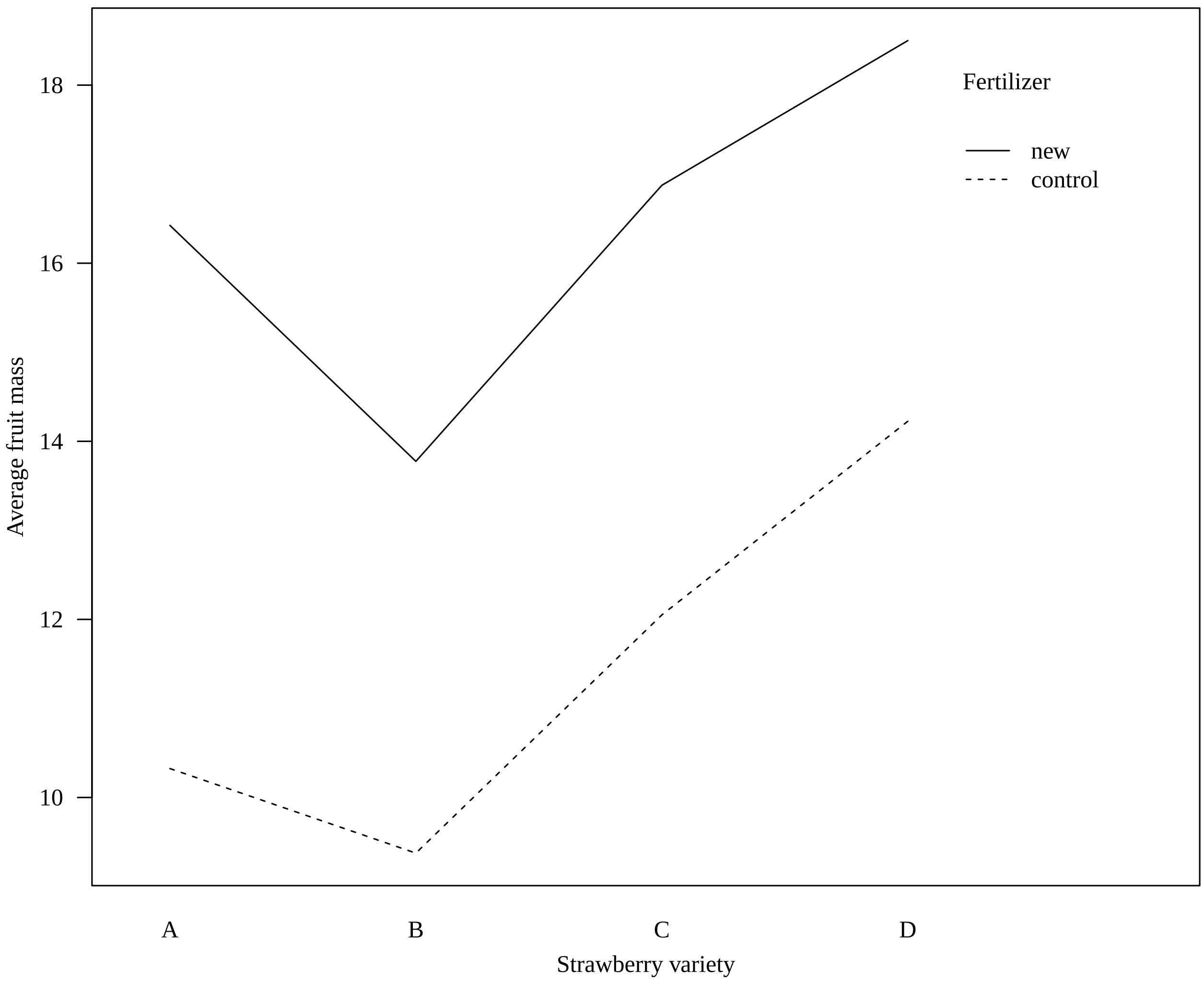
This is one reason why mixed-effects models are needed for split-plot designs.

Interaction Plot

An interaction plot helps visualize whether the effect of fertilizer changes across strawberry varieties.

```r
# Interaction plot
# x-axis:
#   strawberry variety
# separate lines:
#   fertilizer schemes
```

```r
# response:
#   average fruit mass
par(mar = c(3.5, 3.5, 1, 0.5), mgp = c(2.2, 1, 0), las = 1, family = "serif")

with(dat, interaction.plot(
       x.factor = variety,
       trace.factor = fertilizer,
       response = mass,
       xlab = "Strawberry variety",
       ylab = "Average fruit mass",
       trace.label = "Fertilizer"
     ))
```



Interpretation

If the lines are approximately parallel, the fertilizer effect is relatively consistent across varieties.

If the lines are noticeably non-parallel or cross, this suggests a fertilizer-by-variety interaction.

Why Not Use a Standard Two-Way ANOVA?

A standard two-way ANOVA assumes all observations are independent and share the same error structure.

However, in a split-plot design:

- observations within the same whole plot are correlated,
- and the whole-plot factor has fewer independent experimental units.

Ignoring this structure can lead to:

- incorrect standard errors,
- incorrect F-tests,
- and misleading conclusions.

Mixed-effects models properly account for the hierarchical randomization structure.

Takeaway

Split-plot designs arise naturally when some factors are more difficult to randomize than others. The key ideas are:

- different factors may correspond to different experimental-unit levels,
- observations within whole plots are correlated,
- and mixed-effects models are often required to correctly analyze the data.